Data Analytics Project Plan

Predicting Hearing Loss from Routine Health Data (NHANES 2015–2016)

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1. Introduction

Hearing loss is one of the most common sensory impairments worldwide, yet it frequently goes undiagnosed because testing requires specialized equipment and a clinic visit. At the same time, many of the known risk factors for hearing loss — age, noise exposure history, chronic conditions like diabetes, and basic vitals — are already collected during routine health surveys and primary-care visits.

This project explores whether a predictive model can flag individuals at risk of hearing loss using only these general health indicators, **without a hearing test**. If such a model works well enough, it could be used as a low-cost screening tool: clinics could run the model on data they already collect and refer high-risk patients for a full audiological evaluation.

The dataset used is from the CDC's **National Health and Nutrition Examination Survey (NHANES), 2015–2016 cycle** — a large, publicly available, nationally representative survey that includes both the risk-factor data and a gold-standard audiometry exam (the hearing test results used to define our outcome variable).

The project follows the **data science pipeline**: data sourcing → ingestion → engineering → analytics/modeling → visualization → deliverables.

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2. Data Source

Source: CDC NHANES 2015–2016, publicly available at wwwn.cdc.gov/nchs/nhanes

Format: Eight SAS transport files (`.XPT`), covering demographics, audiometry exams, audiological questionnaires, body measures, blood pressure, blood lead/cadmium, medical conditions, and diabetes status.

Size: 4,309 participants (adults aged 20–69 who completed the audiometry exam), 21 columns after feature construction.

Sample Data (first 5 rows of the master table)

seqn	sex	age	military	education	income_ratio	firearm_protection	loud_noise	bmi	blood_pressure	diabetes	hl_presence
83732	Male	62	No	College+	4.39	Gun ≤50% prot.	No Exposure	27.8	120	Yes	0 (No)
83733	Male	53	No	High School	1.32	No Gun	No Exposure	30.8	137	No	1 (Yes)
83735	Female	56	No	College+	5.00	No Gun	No Exposure	42.4	135	No	1 (Yes)
83736	Female	42	No	College+	1.23	No Gun	No Exposure	20.3	106	No	0 (No)
83741	Male	22	No	College+	2.08	No Gun	No Exposure	28.0	112	No	0 (No)

Data Dictionary (key features)

Feature	Type	Description
age	Continuous	Age in years (20–69)
sex	Binary	Male / Female
education	Ordinal (1–3)	Below HS / HS / College+
income_poverty_ratio	Continuous	Family income ÷ poverty line
loud_noise	Ordinal (1–3)	Occupational noise exposure level
very_loud_noise	Ordinal (1–3)	Very loud occupational noise level
firearm_and_protection	Ordinal (1–3)	Firearm use and hearing protection
bmi	Continuous	Body Mass Index
blood_pressure	Continuous	Mean systolic BP (mmHg)
diabetes	Binary	Doctor-diagnosed diabetes

Feature	Type	Description
heart_disease, stroke, cancer	Binary	Self-reported conditions
hl_presence (target)	**Binary**	**Hearing loss present (worse-ear PTA > 25 dB)**

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3. Pipeline Stages

Stage 1 — Data Ingestion

Estimated time: 3 hours

What was done: Downloaded the eight raw `.XPT` files from the CDC servers using an automated Python script, then loaded every file into a local **PostgreSQL** database (containerized with Docker) as its own table. This made the raw data queryable with SQL and kept the ingestion fully reproducible.

Tools: Python (`urllib`, `pyreadstat`, `SQLAlchemy`), PostgreSQL, Docker

Anticipated result — project pipeline:

```
CDC Servers (.XPT files)
        │  Python download script
        ▼
    data/raw/*.XPT
        │  Python loader script
        ▼
    PostgreSQL Database (8 raw tables)
        │  SQL join query
        ▼
    data/processed/raw_master_table.csv  (4,309 rows × 21 cols)
```

Stage 2 — Data Engineering (EDA & Cleaning)

Estimated time: 15 hours

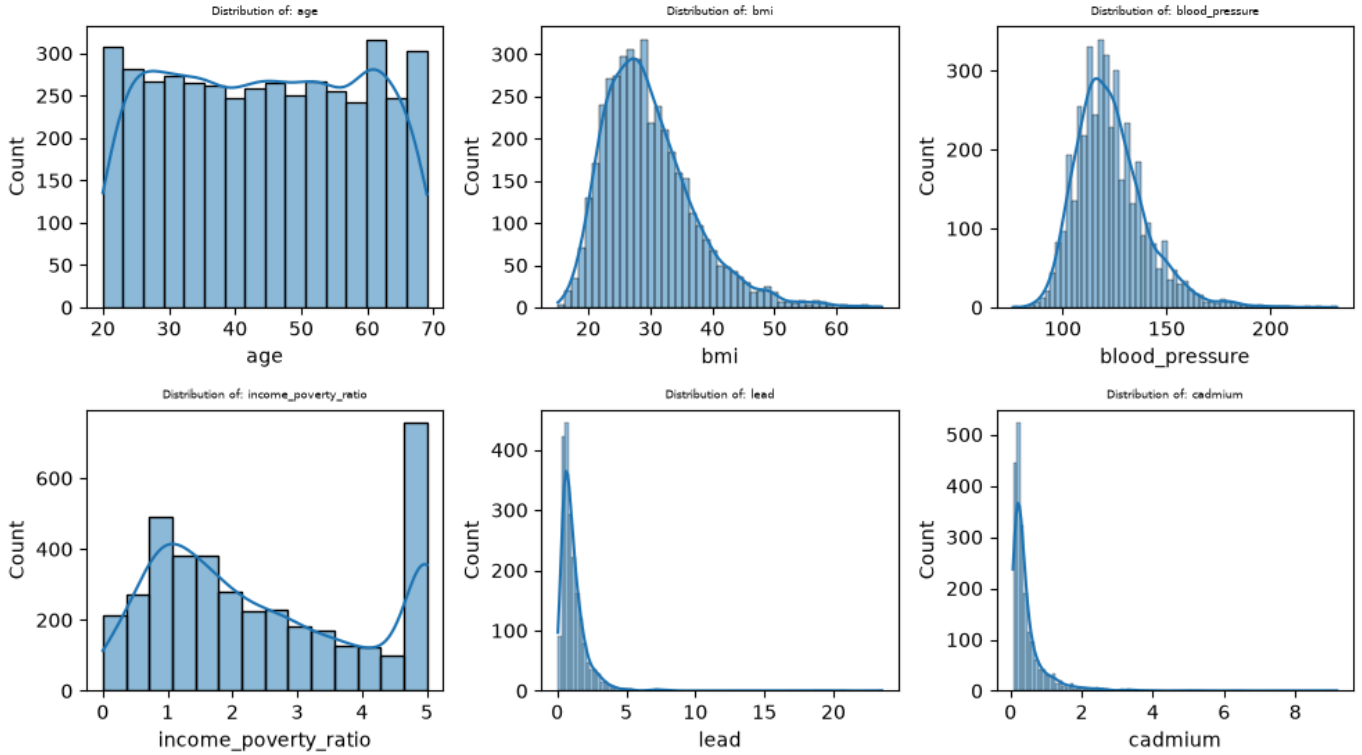
This was the most time-intensive stage. The work included exploratory data analysis (EDA) to understand distributions and patterns, handling missing data, and preparing the dataset for modeling.

Understanding Distributions

The first step was to visualize the distribution of every feature to understand what we are working with — skewness, outliers, class balance, and general data quality.

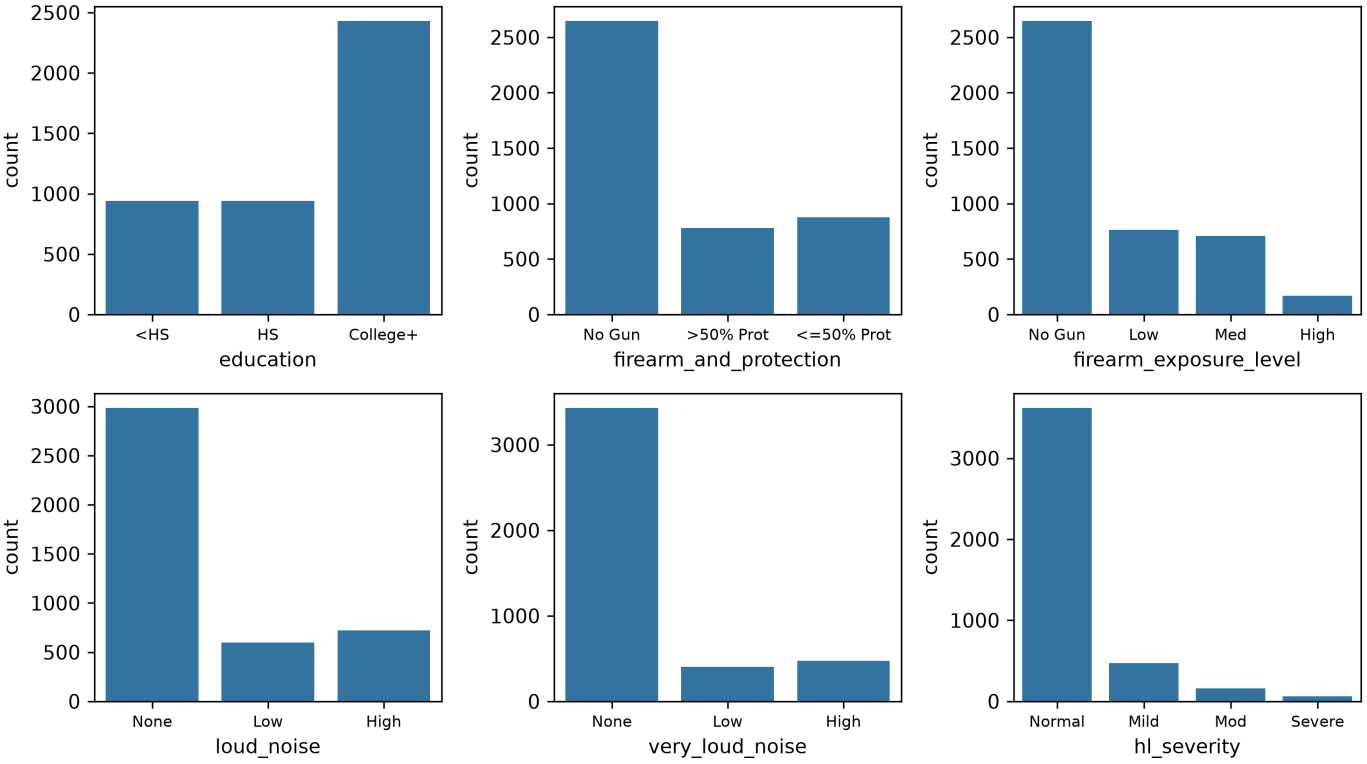
Continuous features (age, BMI, blood pressure, income-to-poverty ratio):

Figure 2 - Distribution of Continuous Features

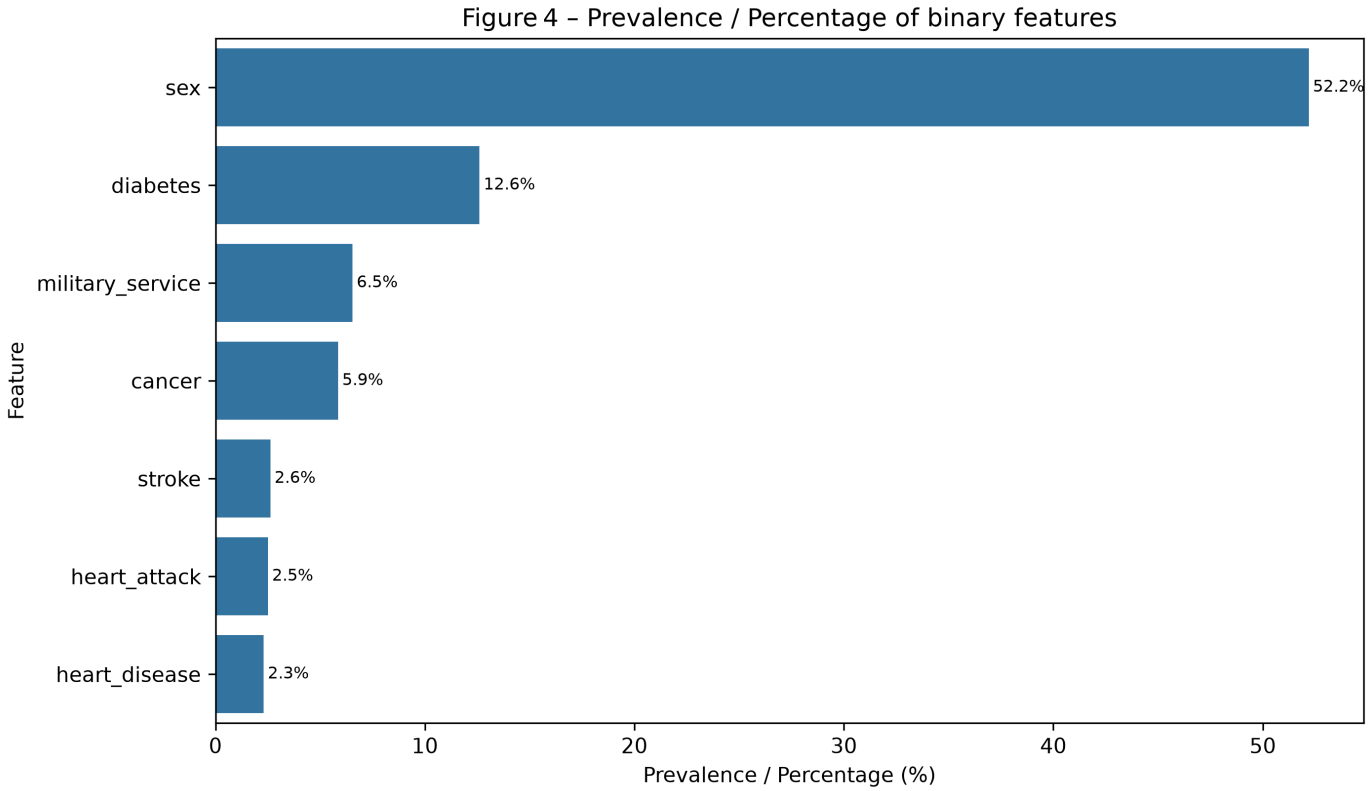


Ordinal features (noise exposure levels, firearm exposure, education):

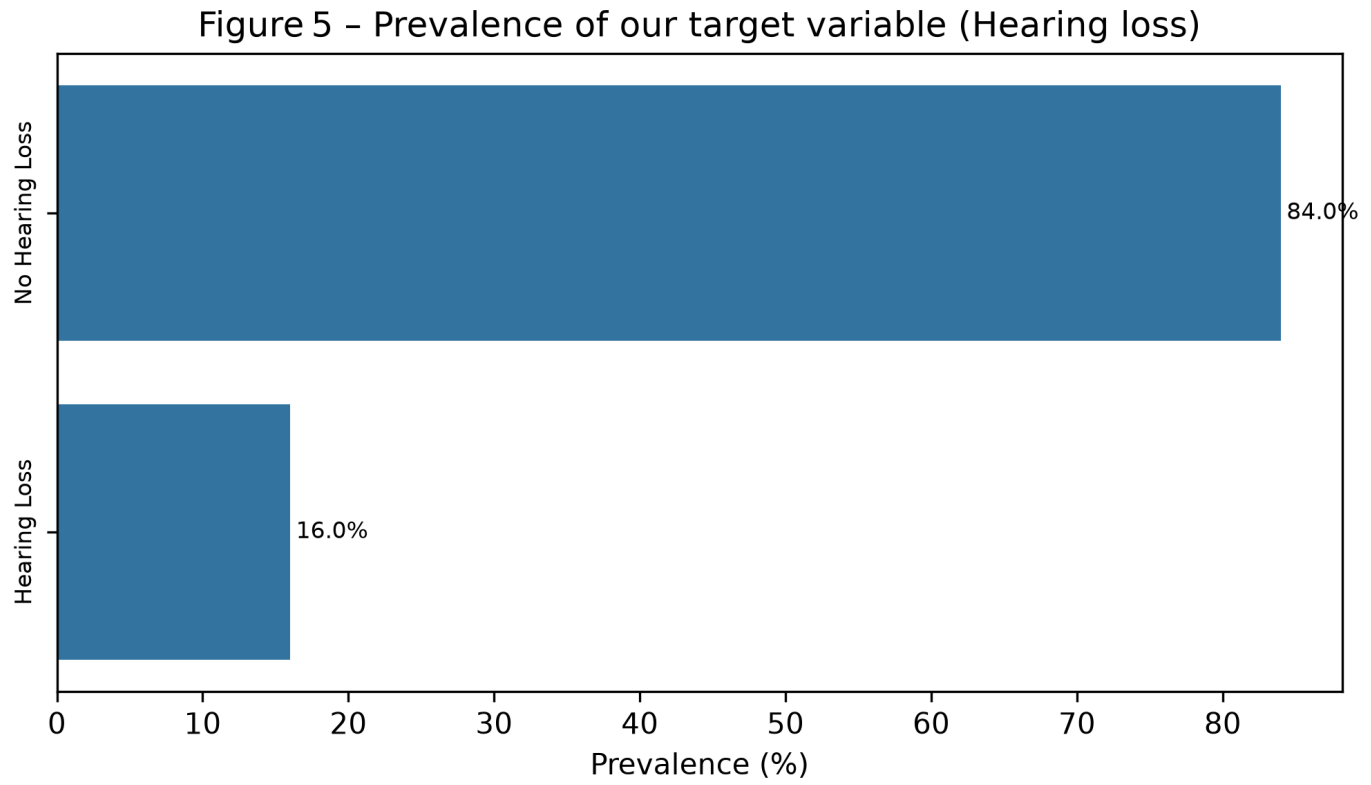
Figure 3 - Distribution of Ordinal Features



Binary features (sex, diabetes, heart disease, etc.):



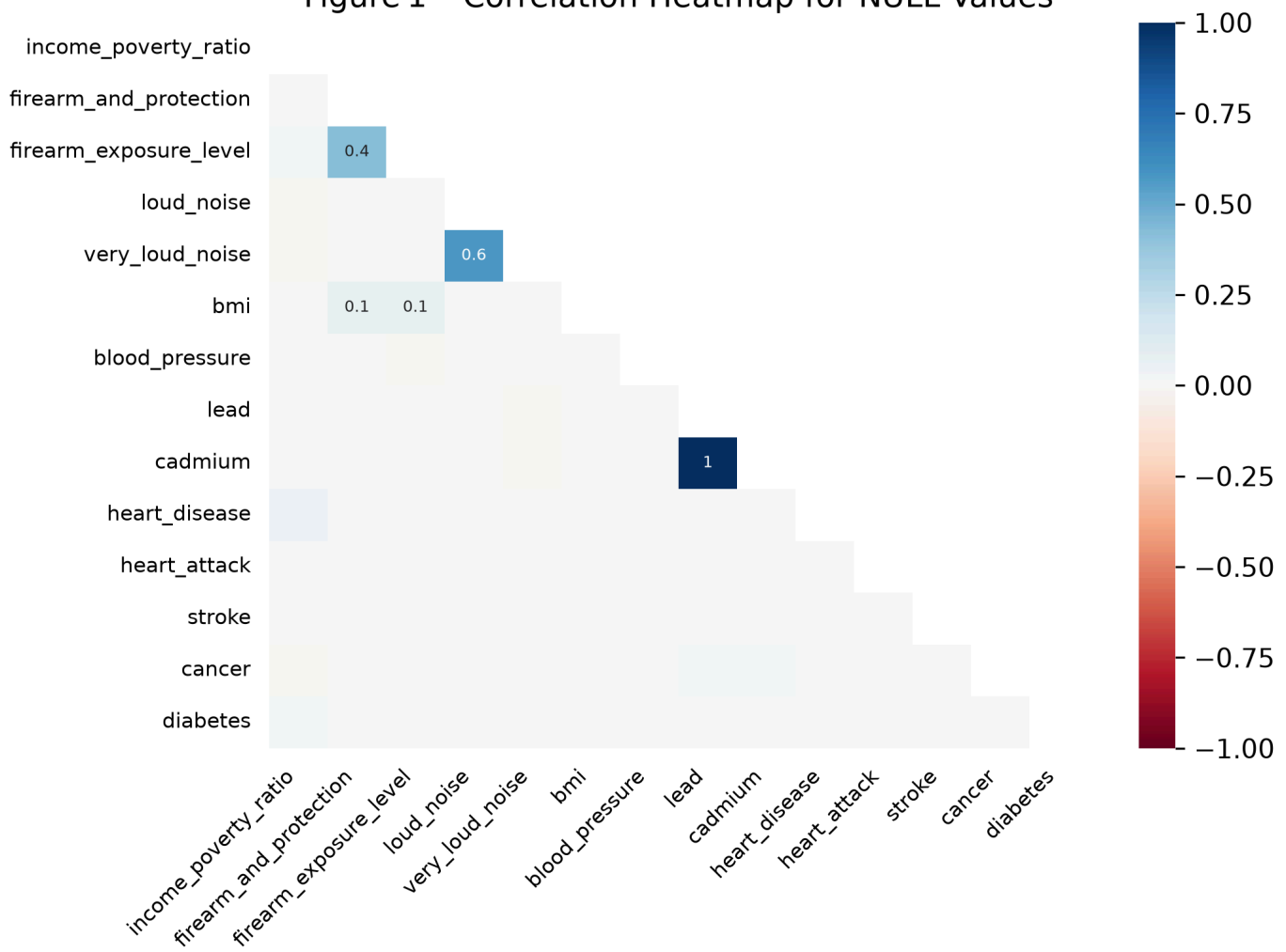
Target variable distribution — hearing loss is present in only ~16% of the sample, creating a class imbalance that had to be addressed during modeling:



Understanding Missingness

NHANES has structured missing data due to its survey skip-logic (e.g., the "very loud noise at work" question was only asked to people who already said they experienced loud noise). A correlation heatmap of NULL values was used to understand which columns had related missingness:

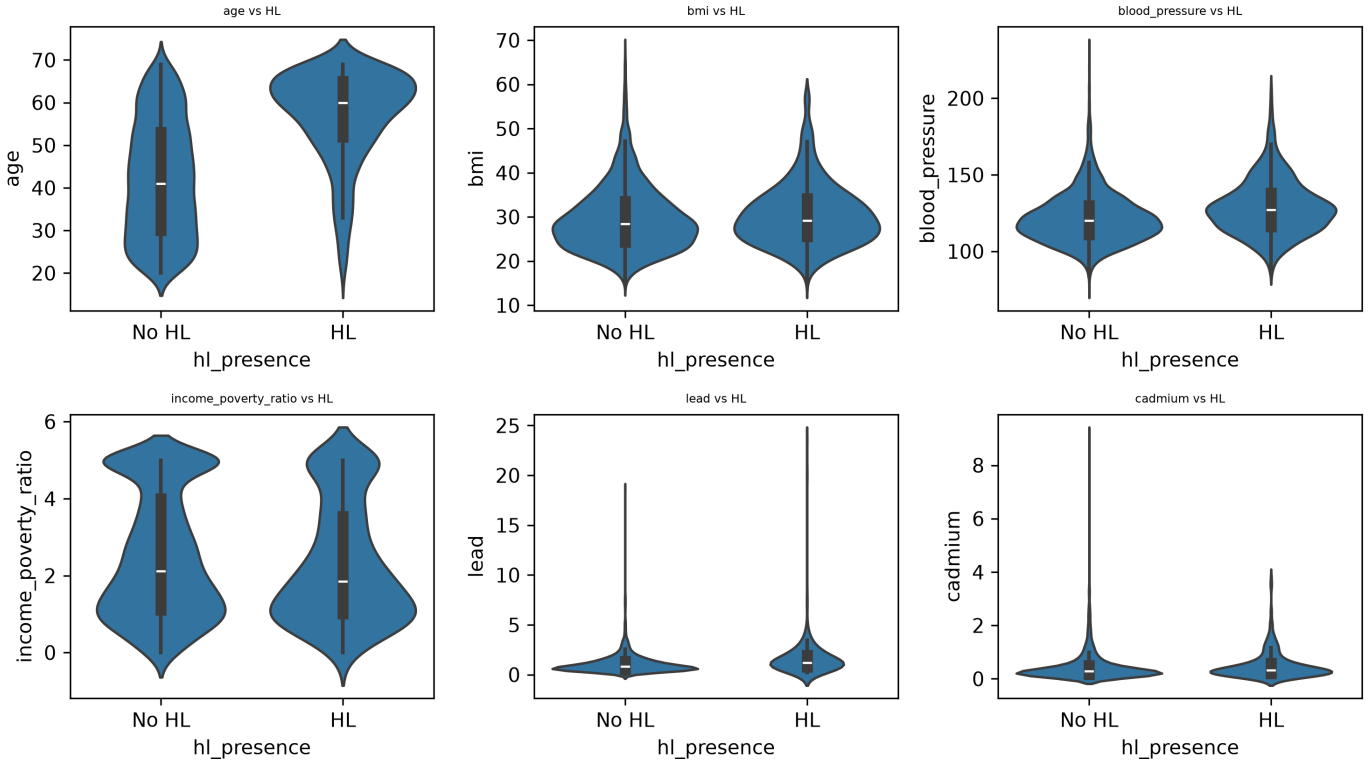
Figure 1 – Correlation Heatmap for NULL Values



Exploring Relationships with Hearing Loss

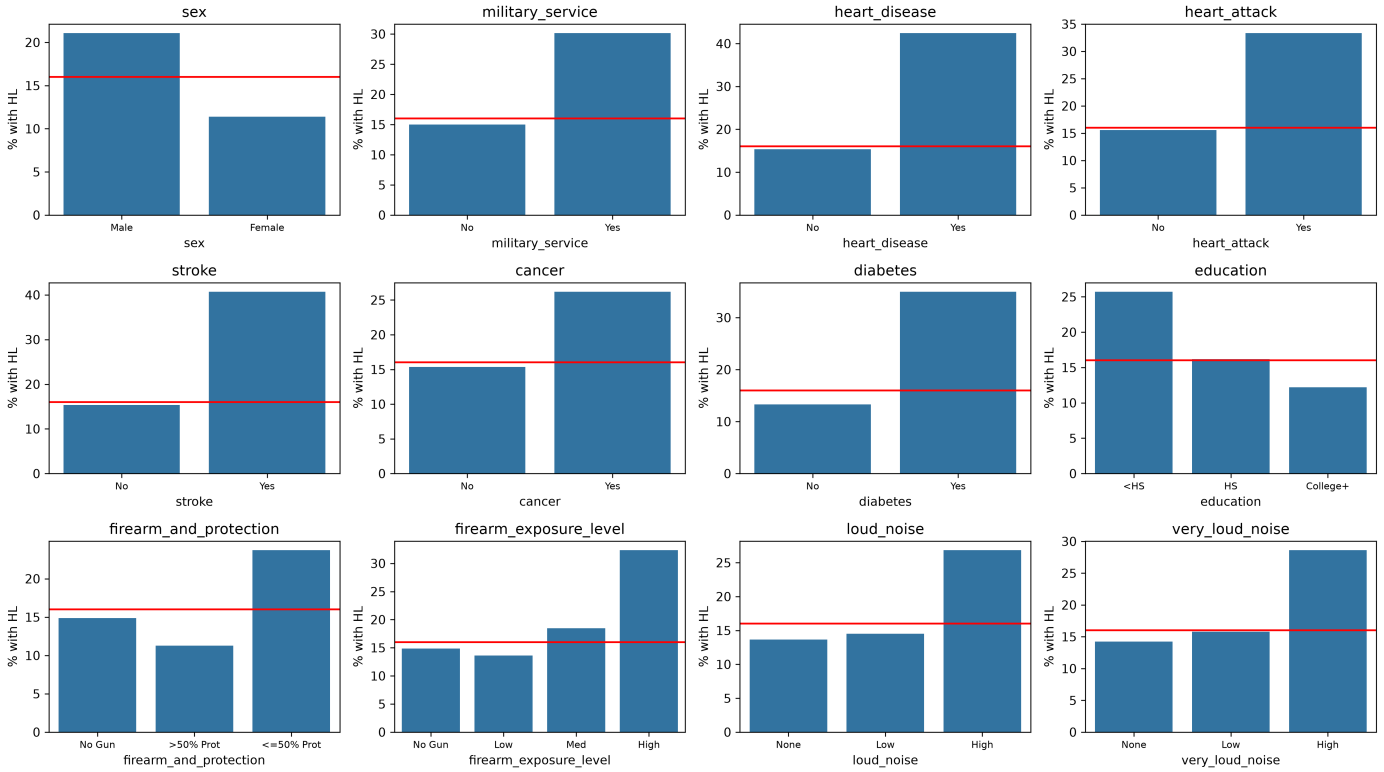
Continuous features plotted against the target variable:

Figure 6 – Continuous features against HL



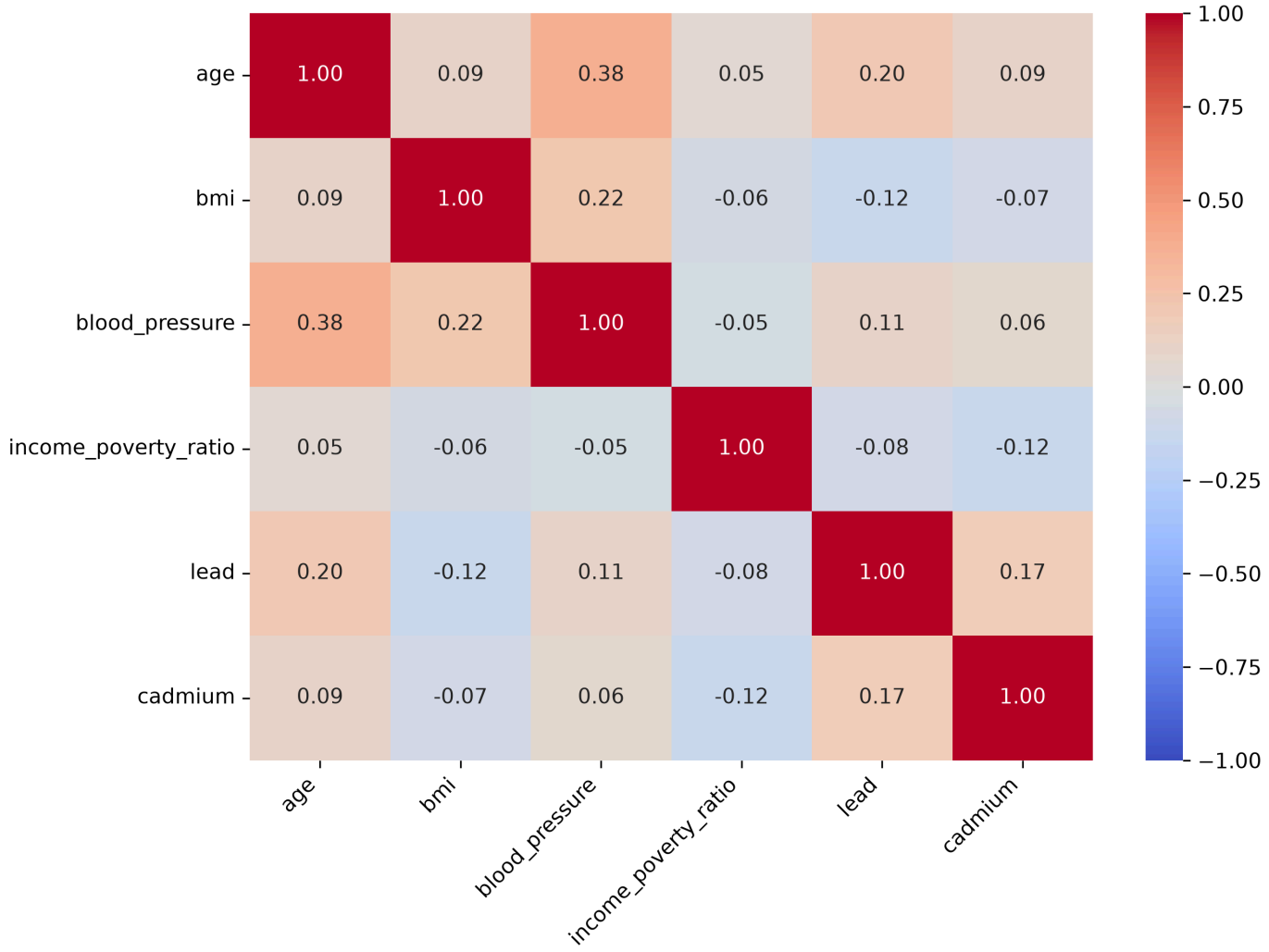
Categorical/ordinal features plotted against the target variable:

Figure 8 - Categorical features against HL



Correlation matrix for continuous features:

Figure 7 - Correlation Matrix for Continuous Features



Cleaning Steps

- Handled skip-logic NULLs with logic-based recoding (not statistical imputation)
- Tested lead & cadmium columns (55% missing) for predictive value — they showed no improvement, so they were dropped
- Imputed remaining missing values appropriately
- Split the data into train/test sets (stratified by the target variable)

Stage 3 — Analytics / Computation (Modeling)

Estimated time: 10 hours

What was done:

- Compared five classifiers (Logistic Regression, Random Forest, XGBoost, KNN, Gaussian Naive Bayes) using 5-fold stratified cross-validation
- Scored models using **PR-AUC** (Precision-Recall Area Under Curve) rather than accuracy, because the target class is imbalanced (~16% positive)
- Tuned the decision threshold to maximize **F2 score** (which weights recall more than precision), since missing a real hearing-loss case is worse than a false alarm
- Logistic Regression performed best, which was consistent with the EDA showing mostly linear relationships between predictors and the target

Results on held-out test set:

Metric	Cross-Validation	Test Set
PR-AUC	0.475	0.454
F2	0.637	0.602
Precision	—	0.329
Recall	—	0.759

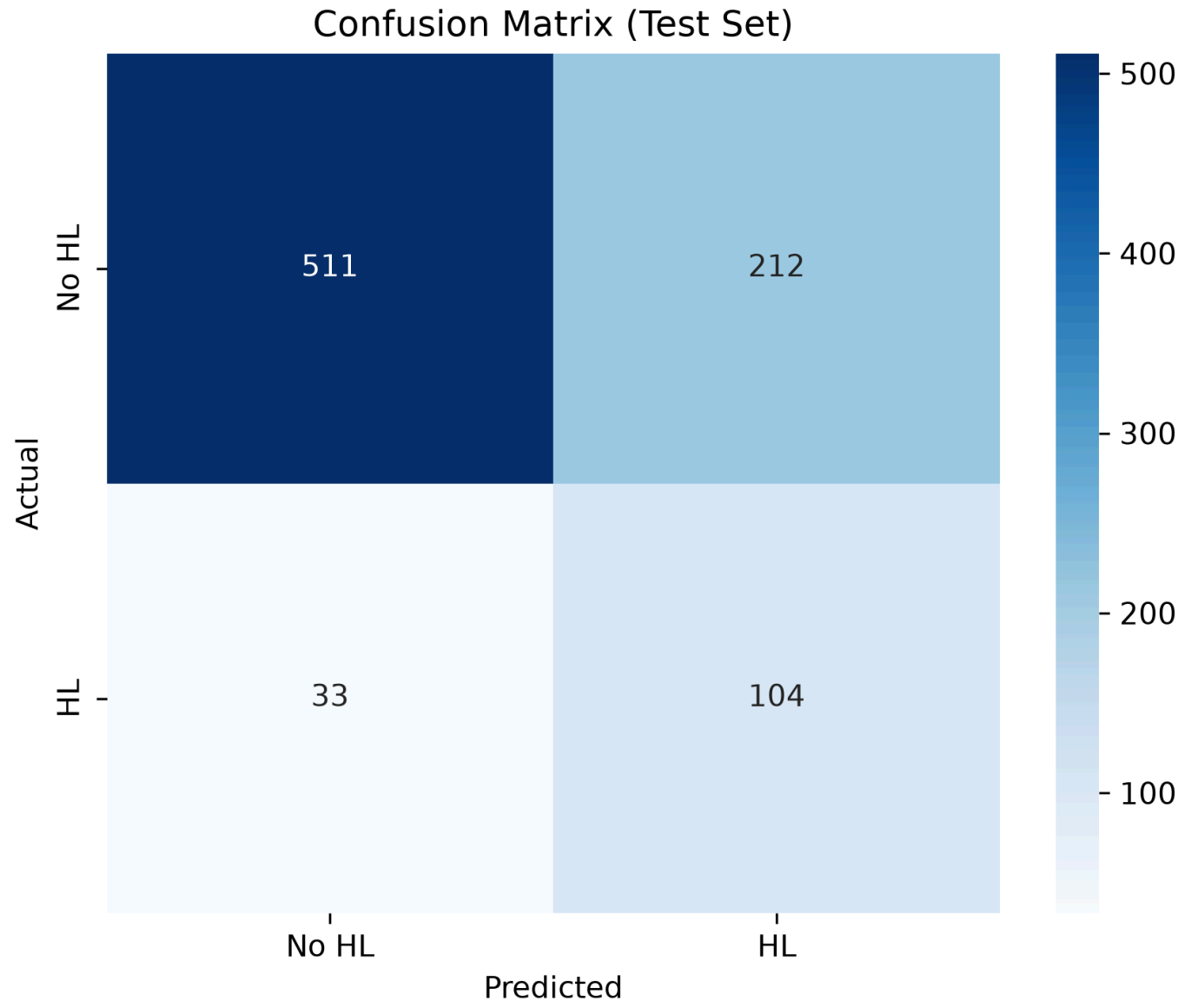
The model catches ~76% of true hearing-loss cases, trading off a higher false-alarm rate — a deliberate choice for a screening tool.

Stage 4 — Visualization (Results)

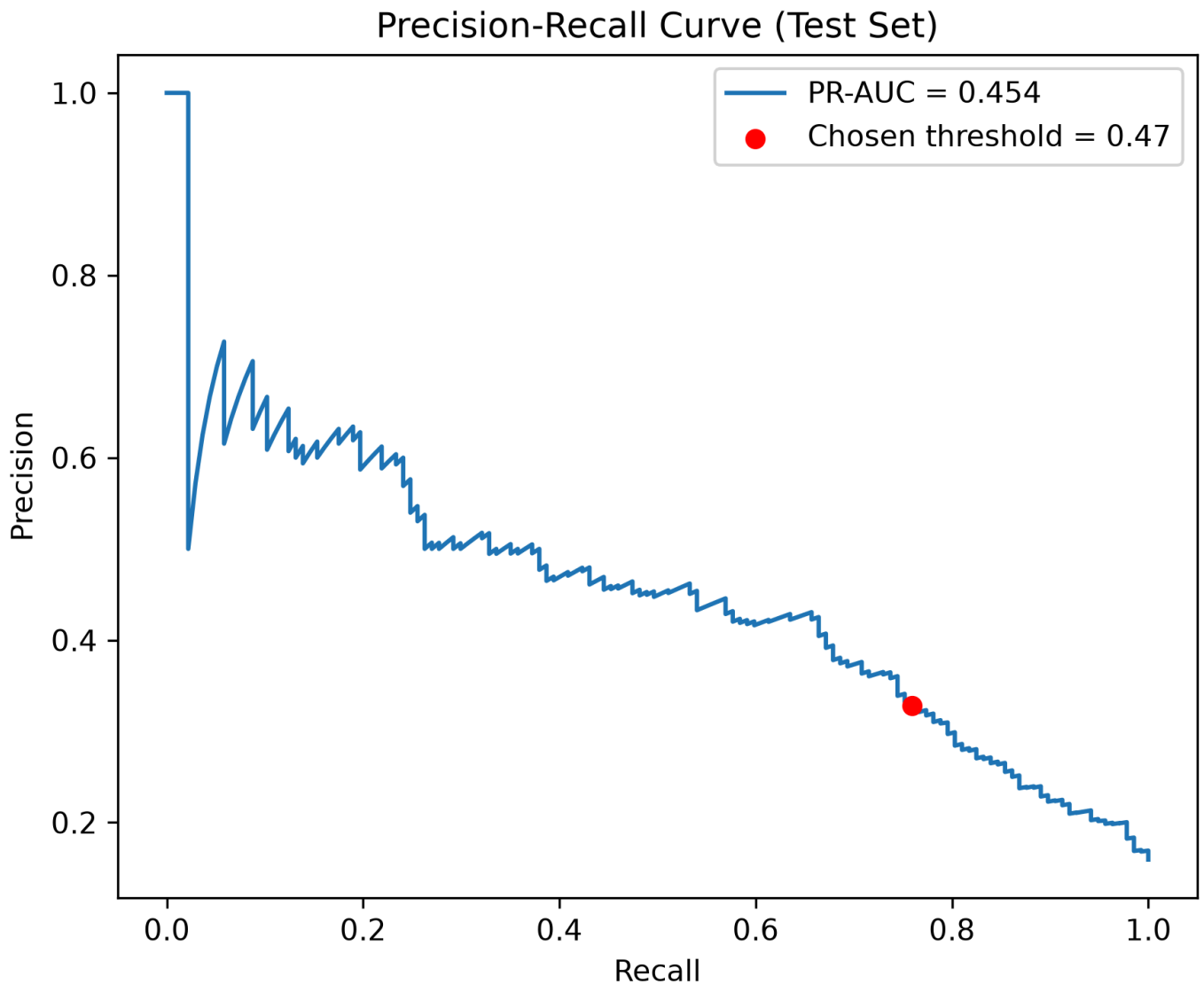
Estimated time: 4 hours

The key visualizations produced for communicating results:

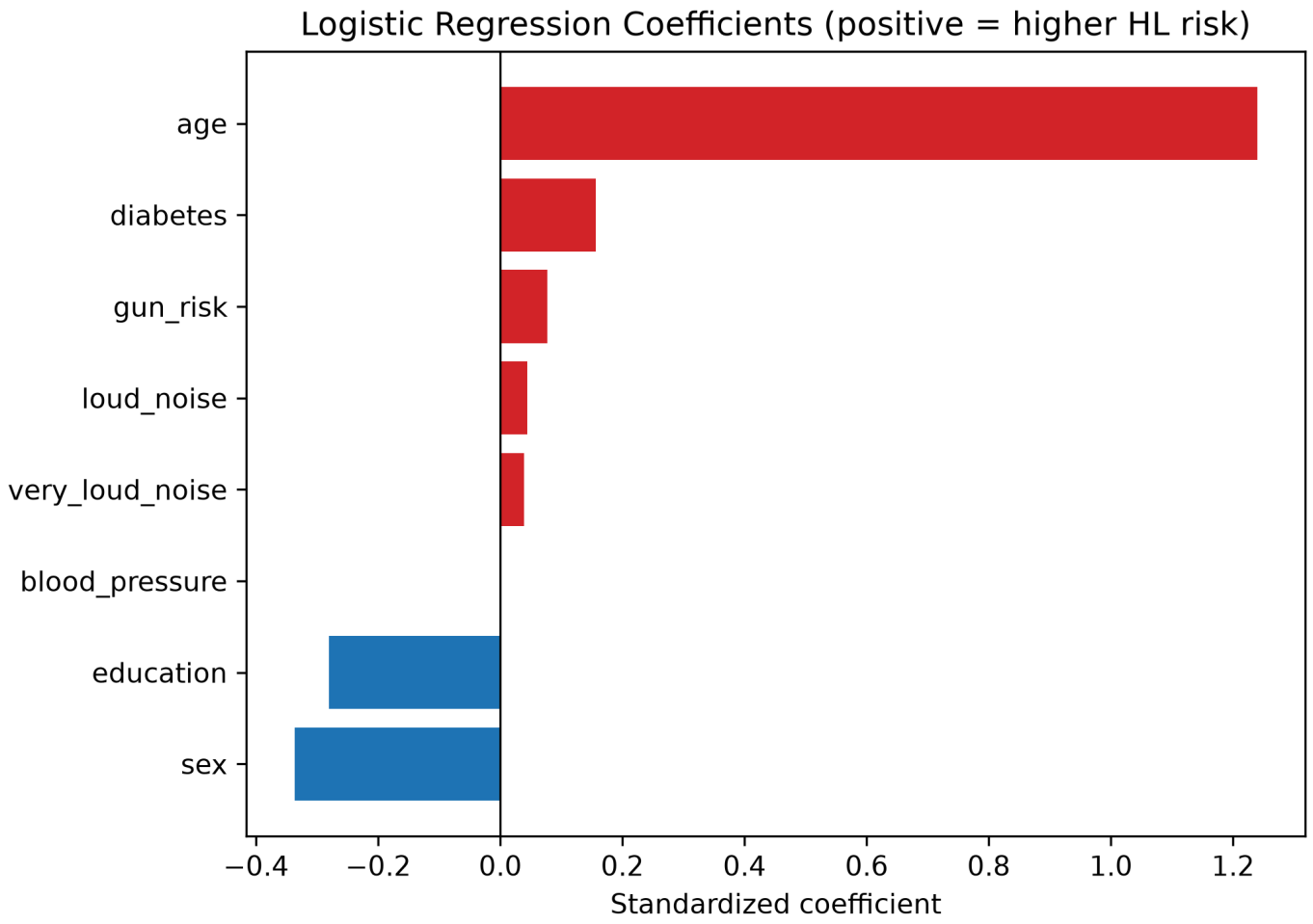
Confusion Matrix — shows the model's predictions vs. actual labels on the test set:



Precision-Recall Curve — illustrates the tradeoff between precision and recall at different thresholds:



Model Coefficients — shows which features the logistic regression model weighted most heavily:



Age is by far the strongest predictor. Diabetes, noise exposure, and being male are associated with higher risk; higher education is associated with lower risk.

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4. Time Estimate Summary

Pipeline Stage	Estimated Hours
Data Ingestion (download, load, SQL joins)	3
Data Engineering (EDA, cleaning, missing data)	15
Analytics / Modeling (training, tuning, evaluation)	10
Visualization (plots, result figures)	4
Deliverables (report writing, packaging)	3
Total	**35**

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5. Deliverables

The following deliverables would be provided to stakeholders:

1. **Written Report (PDF)** — A document summarizing the problem, methodology, findings, and limitations. Includes all key visualizations and a discussion of how the model could be used as a screening tool.
2. **Slide Deck (PDF)** — A concise presentation covering the project motivation, pipeline, key results, and next steps. Suitable for a non-technical audience.
3. **Python Codebase (GitHub Repository)** — The full, reproducible pipeline:
   - `src/` — Data download and database loading scripts
   - `sql/` — SQL query and data dictionary
   - `notebooks/` — EDA, cleaning, and model training notebooks
   - `docker-compose.yml` — One-command database setup
   - `requirements.txt` — All Python dependencies
4. **Trained Model** — The final logistic regression model, ready to generate risk scores on new patient data.

Delivery method: GitHub repository link shared with the team, plus PDF report and slides delivered via email or course submission portal.

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6. Limitations & Future Work

- The model finds associations, not causal relationships (cross-sectional data).
- NHANES survey weights are not applied, so results describe patterns in the sample rather than nationally weighted estimates.
- **Future direction:** Extend the binary model to a multiclass classifier that predicts hearing loss *severity* (mild / moderate / severe) using the `hl_severity` column already constructed in the pipeline.